

AIM : Implement the Adaboost algorithm to create an ensemble of weak classifiers.

Train the ensemble model on a given dataset and evaluate its performance.

Compare the results with individual weak classifiers.

Code :

```
ai p6 T117.py - C:\Users\Adarsh\OneDrive\Videos\ai p6 T117.py (3.12.6)
File Edit Format Run Options Window Help
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import AdaBoostClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, ConfusionMatrixDisplay

print("Adarsh Shukla T117")

df = pd.read_csv(r"C:\Users\Adarsh\Downloads\IRIS.csv")

print("\nDataset:")
print(df.head())

if "species" in df.columns:
    target = "species"
elif "Species" in df.columns:
    target = "Species"
else:
    target = df.columns[-1]

X = df.drop(columns=[target])
y = df[target]

X = X.select_dtypes(include=["number"])

encoder = LabelEncoder()
y = encoder.fit_transform(y)

X_train, X_test, y_train, y_test = train_test_split(
    X,
```

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```
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X = X.select_dtypes(include=["number"])

encoder = LabelEncoder()
y = encoder.fit_transform(y)

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

weak_classifier = DecisionTreeClassifier(
    max_depth=1,
    random_state=42
)

model = AdaBoostClassifier(
    estimator=weak_classifier,
    n_estimators=50,
    learning_rate=1.0,
    random_state=42
)

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("\nAccuracy:", accuracy)

Ln: 81 Col: 0

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model.fit(X_train, y_train)

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("\nAccuracy:", accuracy)

cm = confusion_matrix(y_test, y_pred)

ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=encoder.classes_
).plot()

plt.title("AdaBoost Confusion Matrix")
plt.show()

errors = []

for prediction in model.staged_predict(X_test):
    error = 1 - accuracy_score(y_test, prediction)
    errors.append(error)

plt.plot(range(1, len(errors) + 1), errors)
plt.xlabel("Number of Weak Classifiers")
plt.ylabel("Error Rate")
plt.title("AdaBoost Error Rate")
plt.grid()
plt.show()

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```

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Output :

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Weak Classifier Accuracy: 66.66666666666666 %

AdaBoost Accuracy: 93.33333333333333 %

```
IDLE Shell 3.12.6
File Edit Shell Debug Options Window Help
Python 3.12.6 (tags/v3.12.6:a4a2d2b, Sep 6 2024, 20:11:23) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/Adarsh/OneDrive/Videos/ai p6 T117.py
Adarsh Shukla T117

Dataset:
  sepal_length  sepal_width  petal_length  petal_width  species
0          5.1           3.5           1.4           0.2  Iris-setosa
1          4.9           3.0           1.4           0.2  Iris-setosa
2          4.7           3.2           1.3           0.2  Iris-setosa
3          4.6           3.1           1.5           0.2  Iris-setosa
4          5.0           3.6           1.4           0.2  Iris-setosa

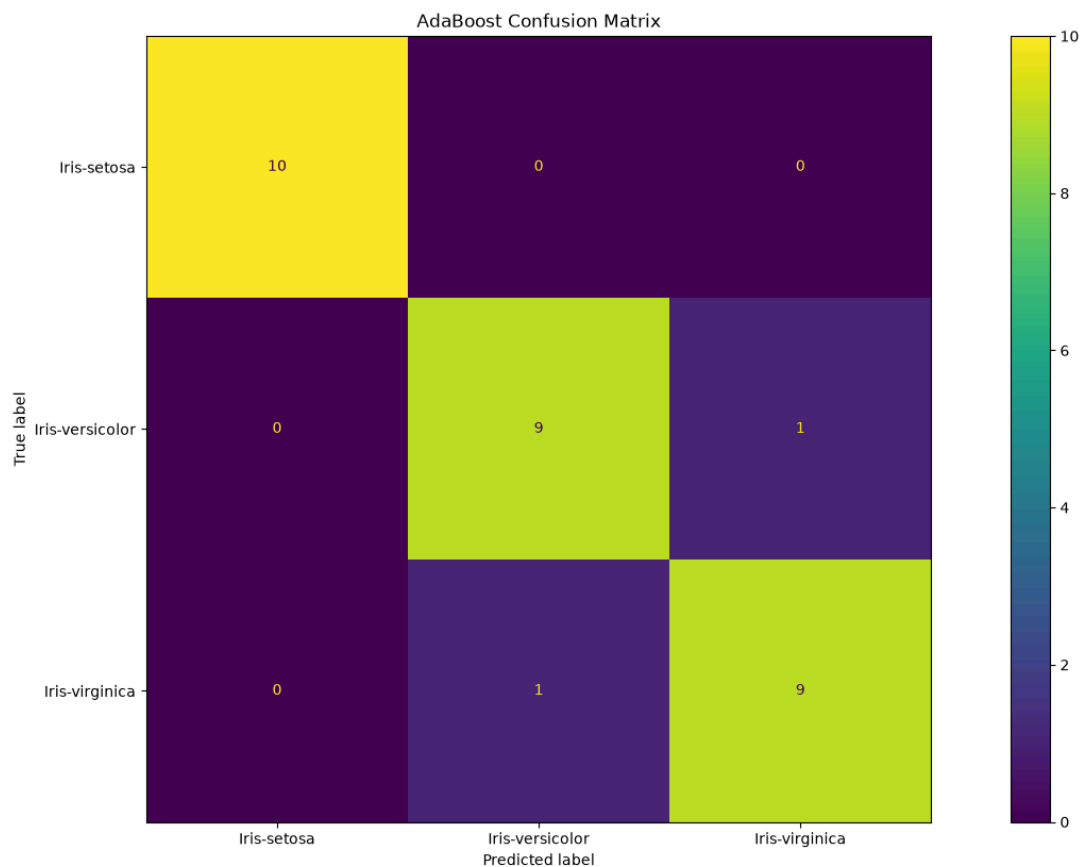
Accuracy: 0.9333333333333333
```

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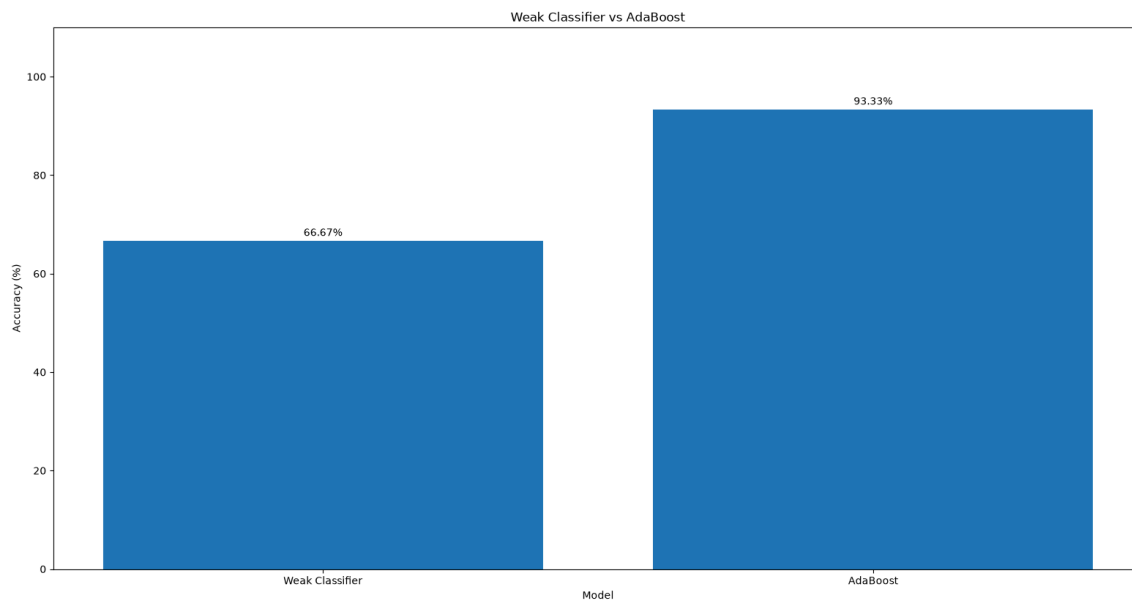
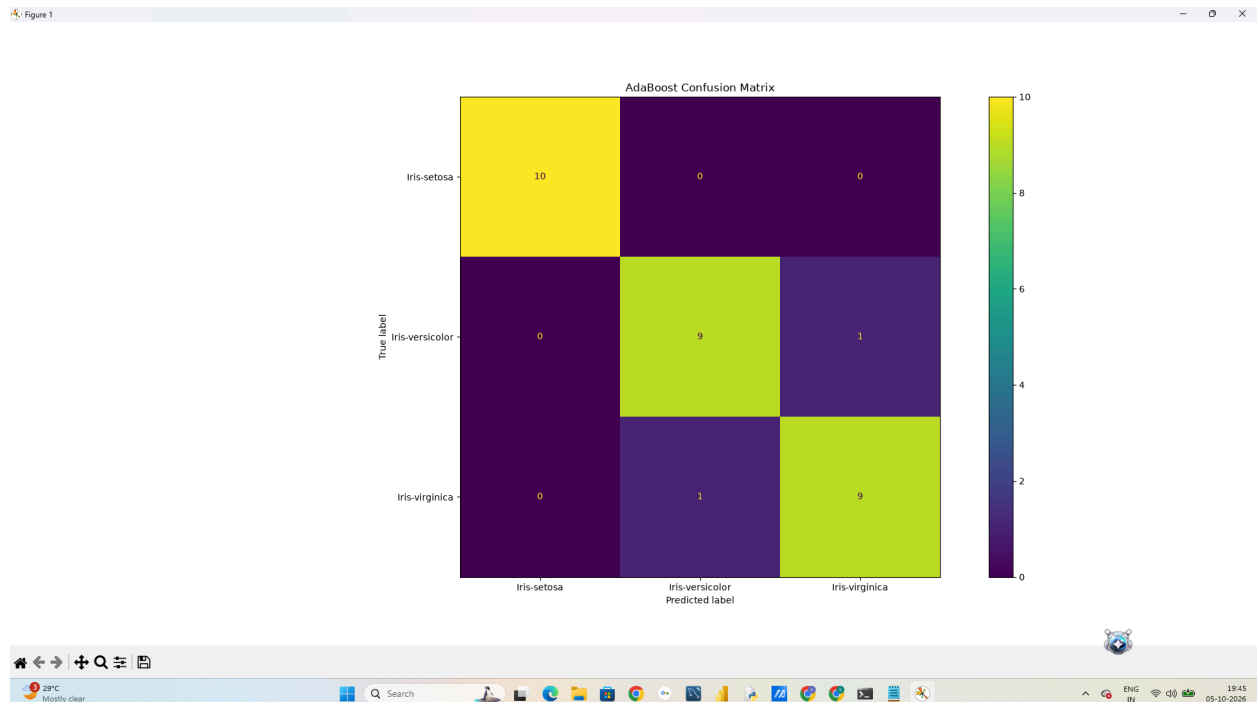


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